Screenshots of outputs

Count of images in each class folder

```
#count images inside the folders
for cls in classes:
    cls_path = os.path.join(DATA_DIR, cls)
    count = len([f for f in os.listdir(cls_path) if f.lower().endswith('.jpg')])
    print(f'  {cls}: {count} images')
```

```
  algal leaf: 113 images
  brown blight: 113 images
  white spot: 142 images
```

Train,val and test data counts

```
Train: 257, Val: 55, Test: 56
```

VGG16 Model

Check number of trainable parameters

```
#Check number of trainable parameters
trainable = sum(p.numel() for p in model_vgg16.parameters() if p.requires_grad)
total = sum(p.numel() for p in model_vgg16.parameters())

print(f'VGG16 — trainable params: {trainable:,} / {total:,}')
```

```
VGG16 — trainable params: 119,558,147 / 134,272,835
```

Train VGG16

```
vgg_train_loss, vgg_val_loss, vgg_train_acc, vgg_val_acc = train_model(
    model_vgg16,
    train_loader,
    val_loader,
    optimizer_vgg16,
    epochs=20,
    patience=3
)
```

```
Epoch 1
Train Loss: 17.6634 | Train Acc: 0.4202
Val Loss: 3.4243 | Val Acc: 0.6727
Best model saved

Epoch 2
Train Loss: 15.3519 | Train Acc: 0.5409
Val Loss: 3.5606 | Val Acc: 0.5636
No improvement (1/3)

Epoch 3
Train Loss: 15.9169 | Train Acc: 0.5642
Val Loss: 3.5473 | Val Acc: 0.5818
No improvement (2/3)

Epoch 4
Train Loss: 13.9330 | Train Acc: 0.6265
Val Loss: 3.1887 | Val Acc: 0.7091
Best model saved

Epoch 5
Train Loss: 12.4192 | Train Acc: 0.6926
Val Loss: 2.5084 | Val Acc: 0.7455
Best model saved

Epoch 6
Train Loss: 14.2814 | Train Acc: 0.6187
Val Loss: 2.2613 | Val Acc: 0.8182
Best model saved

Epoch 7
Train Loss: 11.5790 | Train Acc: 0.7043
Val Loss: 2.4354 | Val Acc: 0.7636
No improvement (1/3)

Epoch 8
Train Loss: 12.4132 | Train Acc: 0.6926
Val Loss: 2.6383 | Val Acc: 0.7636
No improvement (2/3)

Epoch 9
Train Loss: 10.6965 | Train Acc: 0.7704
Val Loss: 2.5048 | Val Acc: 0.7818
No improvement (3/3)
Early stopping triggered
```

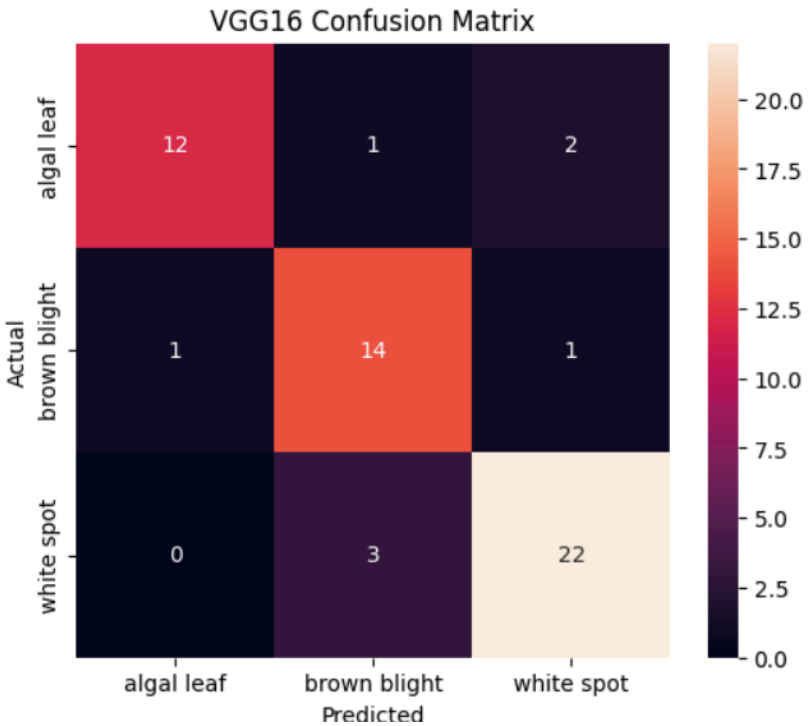
Evaluate VGG16

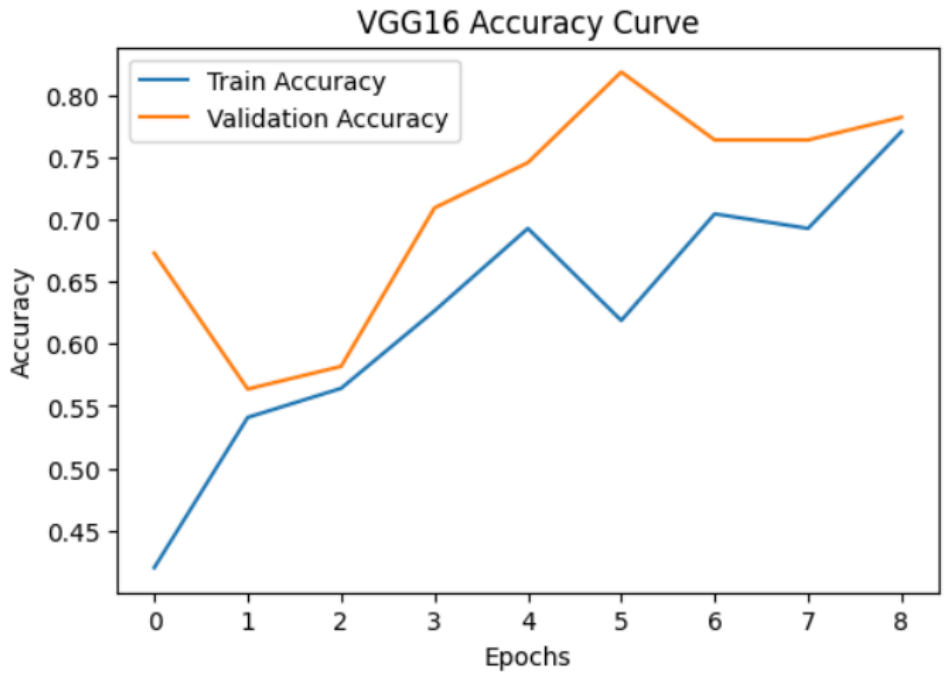
```
vgg_acc, y_true_vgg, y_pred_vgg = evaluate_model(
    model_vgg16,
    test_loader
)

print("VGG16 Test Accuracy:", vgg_acc)
```

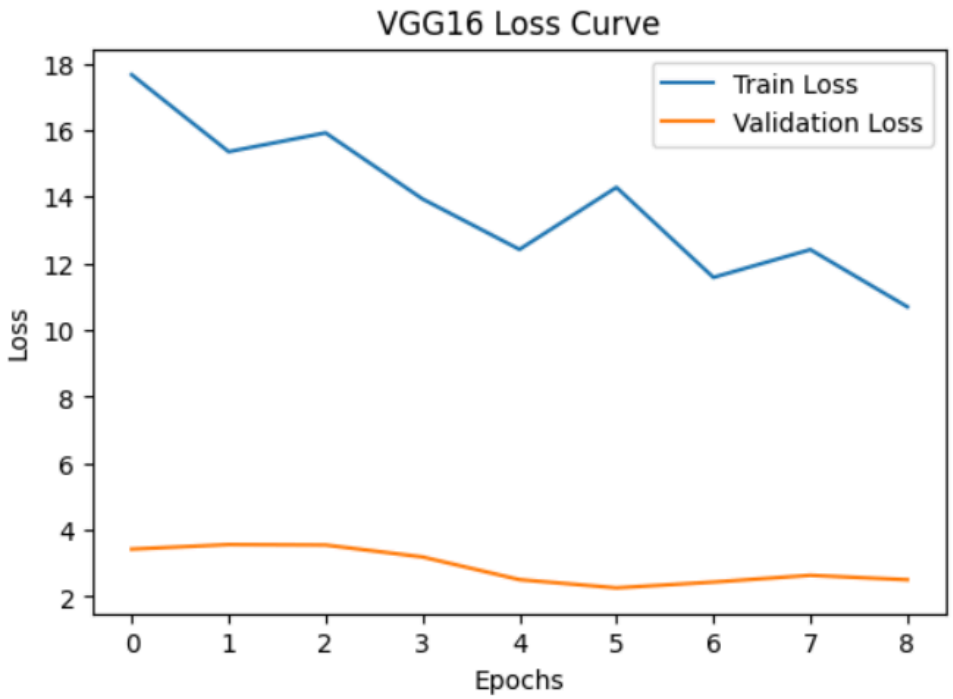
```
VGG16 Test Accuracy: 0.6785714285714286
```

```
#Classification report VGG16
print(classification_report(
    y_true_vgg,
    y_pred_vgg,
    target_names=CLASS_NAMES
))
```

```
              precision    recall  f1-score   support

   algal leaf       0.92      0.80      0.86        15
 brown blight       0.78      0.88      0.82        16
   white spot       0.88      0.88      0.88        25

     accuracy                           0.86        56
    macro avg       0.86      0.85      0.85        56
 weighted avg       0.86      0.86      0.86        56
```







ResNet50 Model

Calculate number of trainable parameters

```
trainable = sum(p.numel() for p in model_resnet50.parameters() if p.requires_grad)
total = sum(p.numel() for p in model_resnet50.parameters())

print(f'ResNet50 — trainable params: {trainable:,} / {total:,}')
```

```
ResNet50 — trainable params: 14,970,883 / 23,514,179
```

Train ResNet50

```
resnet_train_loss, resnet_val_loss, resnet_train_acc, resnet_val_acc = train_model(
    model_resnet50,
    train_loader,
    val_loader,
    optimizer_resnet50,
    epochs=20,
    patience=3
)
```

```
Epoch 1
Train Loss: 18.5868 | Train Acc: 0.4047
Val Loss: 4.2404 | Val Acc: 0.4545
Best model saved

Epoch 2
Train Loss: 16.5357 | Train Acc: 0.6809
Val Loss: 3.7168 | Val Acc: 0.6182
Best model saved

Epoch 3
Train Loss: 13.6204 | Train Acc: 0.7160
Val Loss: 3.3188 | Val Acc: 0.6364
Best model saved

Epoch 4
Train Loss: 10.6625 | Train Acc: 0.7821
Val Loss: 2.7772 | Val Acc: 0.7455
Best model saved

Epoch 5
Train Loss: 7.9704 | Train Acc: 0.8366
Val Loss: 2.5607 | Val Acc: 0.7636
Best model saved

Epoch 6
Train Loss: 7.7693 | Train Acc: 0.8638
Val Loss: 2.2790 | Val Acc: 0.7455
Best model saved

Epoch 7
Train Loss: 4.8231 | Train Acc: 0.9183
Val Loss: 2.3578 | Val Acc: 0.7636
No improvement (1/3)

Epoch 8
Train Loss: 4.1103 | Train Acc: 0.9339
Val Loss: 2.3567 | Val Acc: 0.8000
No improvement (2/3)

Epoch 9
Train Loss: 4.6453 | Train Acc: 0.9300
Val Loss: 2.2310 | Val Acc: 0.7636
Best model saved

Epoch 10
Train Loss: 3.2496 | Train Acc: 0.9455
Val Loss: 2.1254 | Val Acc: 0.8364
Best model saved

Epoch 11
Train Loss: 2.5326 | Train Acc: 0.9650
Val Loss: 2.3293 | Val Acc: 0.8000
No improvement (1/3)

Epoch 12
Train Loss: 2.8330 | Train Acc: 0.9611
Val Loss: 2.5066 | Val Acc: 0.7818
No improvement (2/3)

Epoch 13
Train Loss: 3.4953 | Train Acc: 0.9689
Val Loss: 2.4487 | Val Acc: 0.8182
No improvement (3/3)
Early stopping triggered
```

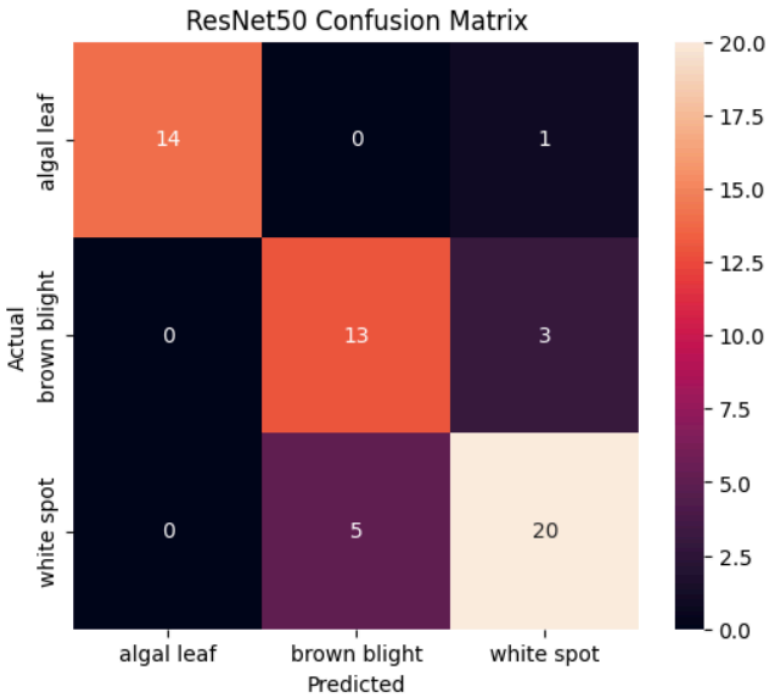
Evaluate ResNet50

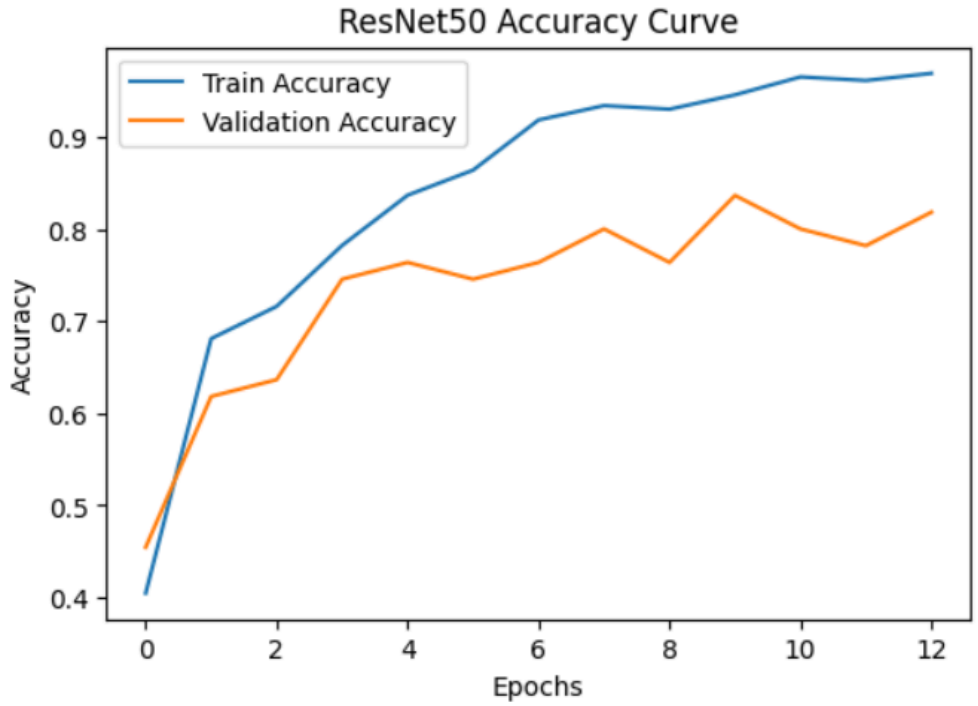
```
resnet_acc, y_true_resnet, y_pred_resnet = evaluate_model(
    model_resnet50,
    test_loader
)

print("ResNet50 Test Accuracy:", resnet_acc)
```

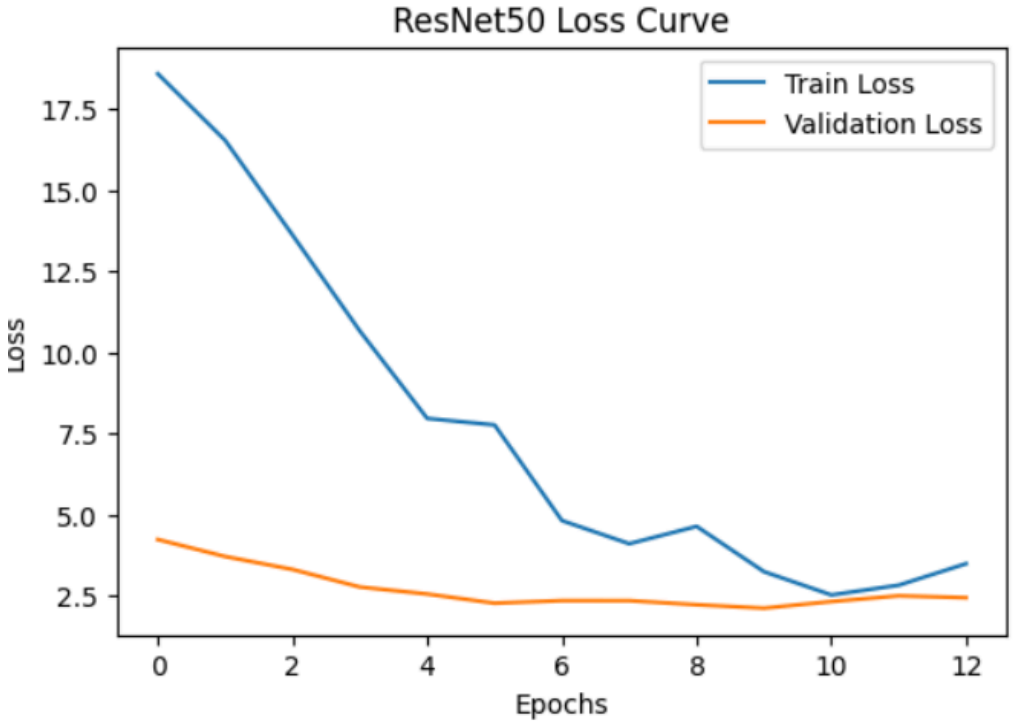
```
C:\Users\Nuwani Hitibandara\AppData\Local\Programs\Python\Python311\Lib\site-pac
gument is set as true but no accelerator is found, then device pinned memory wor
  super().__init__(loader)
ResNet50 Test Accuracy: 0.8392857142857143
```

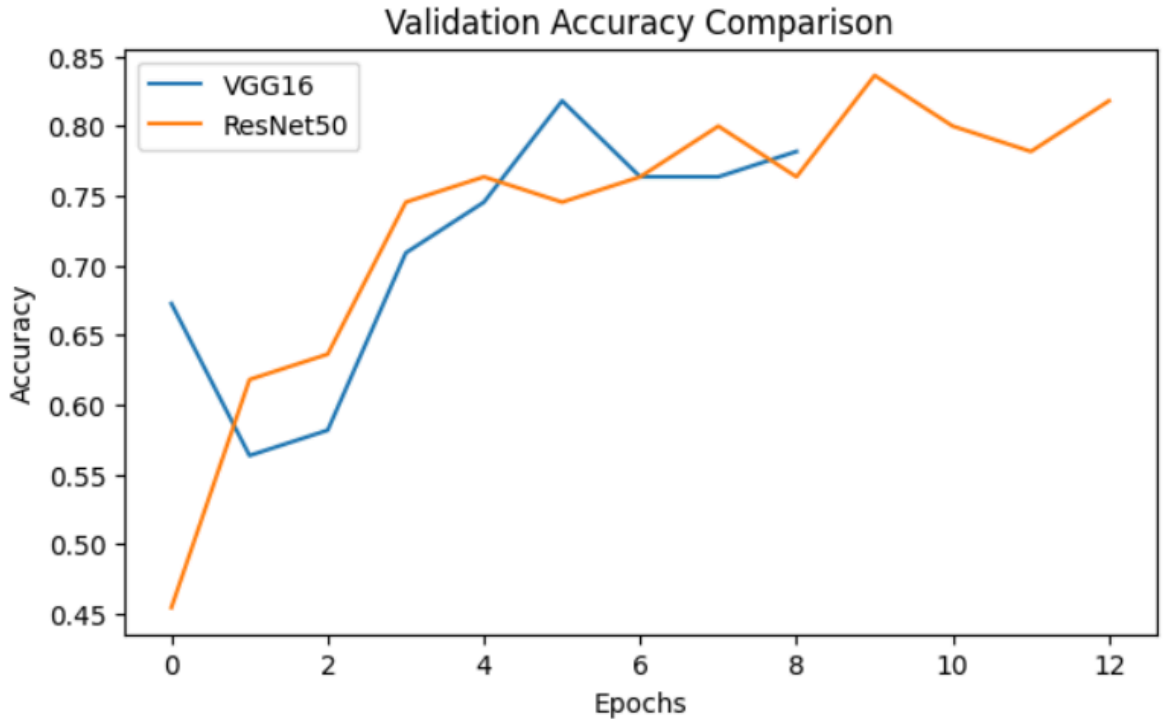
```
#classification report ResNet50
print(classification_report(
    y_true_resnet,
    y_pred_resnet,
    target_names=CLASS_NAMES
))
```

```
              precision    recall  f1-score   support

  algal leaf       1.00      0.93      0.97        15
brown blight       0.72      0.81      0.76        16
  white spot       0.83      0.80      0.82        25

    accuracy                           0.84        56
   macro avg       0.85      0.85      0.85        56
weighted avg       0.85      0.84      0.84        56
```




ResNet50 Accuracy Curve
Train Accuracy
Validation Accuracy
Accuracy
0.4
0.5
0.6
0.7
0.8
0.9
0
2
4
6
8
10
12
Epochs


ResNet50 Loss Curve
Train Loss
Validation Loss
Loss
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
2
4
6
8
10
12
Epochs


Validation Accuracy Comparison
VGG16
ResNet50
Accuracy
Epochs
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0
2
4
6
8
10
12

Validation Loss Comparison
VGG16
ResNet50
Loss
Epochs
4.0
3.5
3.0
2.5
0
2
4
6
8
10
12